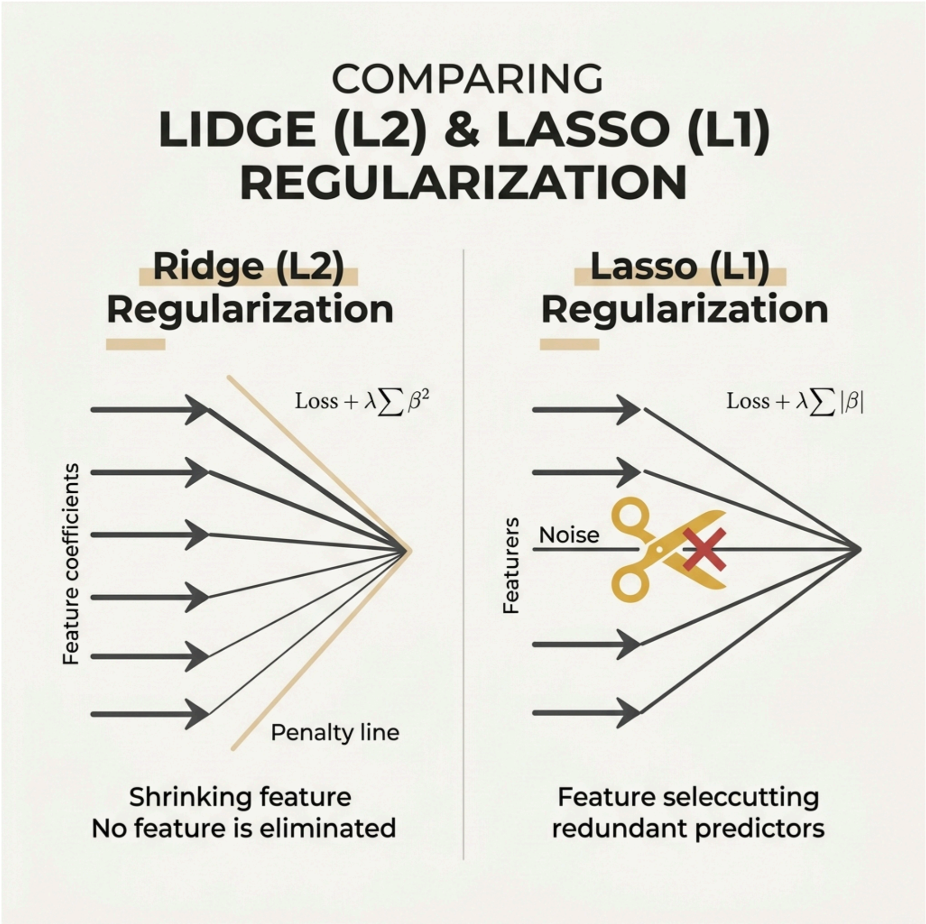
STEP 1: PROBLEM & SOLUTION

Linear Regression in Business & OLS Limitations

Why does OLS fail in practice?

- Multicollinearity: Strongly correlated features cause coefficients to explode, losing stability.
- Overfitting: OLS keeps all features, fitting noise, which degrades generalization on new test data.
- High Dimension (p > N): Analytical solution fails as matrix inversion is impossible.



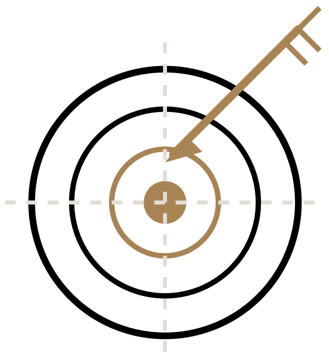
Regularization Solution

Ridge (L2) shrinks coefficients smoothly for stability. Lasso (L1) prunes noise features to exactly 0.0000.

SELECTION CRITERIA

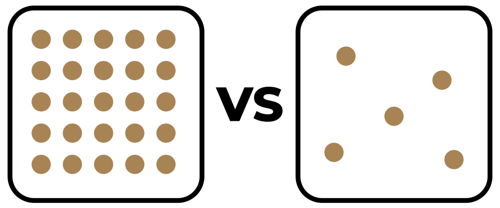
RIDGE vs LASSO Regression

01 MAIN OBJECTIVE



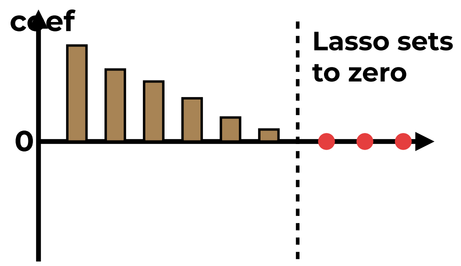
Goal
(Accuracy vs Interpretability)

02 SIGNAL STRUCTURE



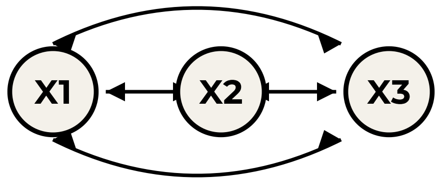
Signal
(Dense vs Sparse)

03 FEATURE SELECTION



Selection
(Ridge keeps all / Lasso drops some)

04 MULTICOLLINEARITY

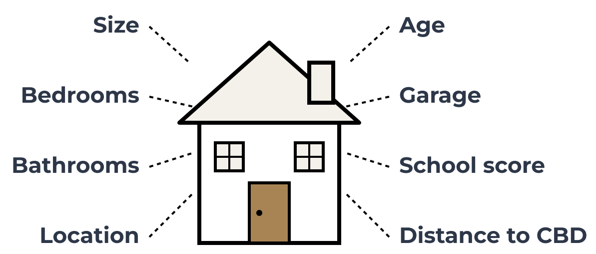


Correlation
(Ridge shares / Lasso picks one)

WHEN TO USE

Real-World Scenarios & Key Limits

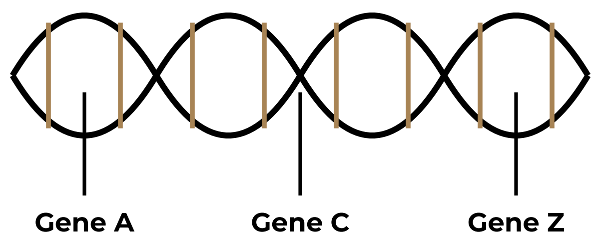
01 DENSE SIGNAL



Use Ridge

(House prices — all features matter)

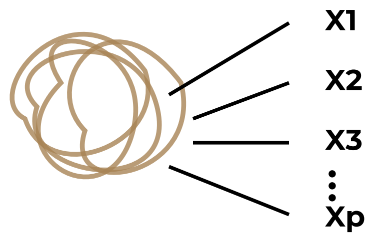
02 SPARSE SIGNAL



Use Lasso

(Gene data — few key genes)

03 HIGH MULTICOLLINEARITY



Ridge stable / Lasso trims

(Lasso caps & trims / Ridge shares weights)

04 KEY LIMITS

Remember

(p > n: Lasso caps at n · Standardize)

DATASET CRITERIA

Algorithm Selection Matching Guidelines

→ When to choose LASSO (L1)

- ✓ Sparse Signal - Many features are irrelevant noise, automatic pruning is needed.
- ✓ High-dimensional settings where features far outnumber samples ($p \gg N$).
- ✓ Interpretability focus - Needs a simple model with few non-zero weights.
- ✓ Independent features - Low grouping correlation or cluster effects.

Real Example: Genomics Target Identification

Study on 300 samples with 20,000 genes. Lasso sets 19,950 noise genes to exactly 0, isolating 50 key target genes for drug synthesis.

→ When to choose RIDGE (L2)

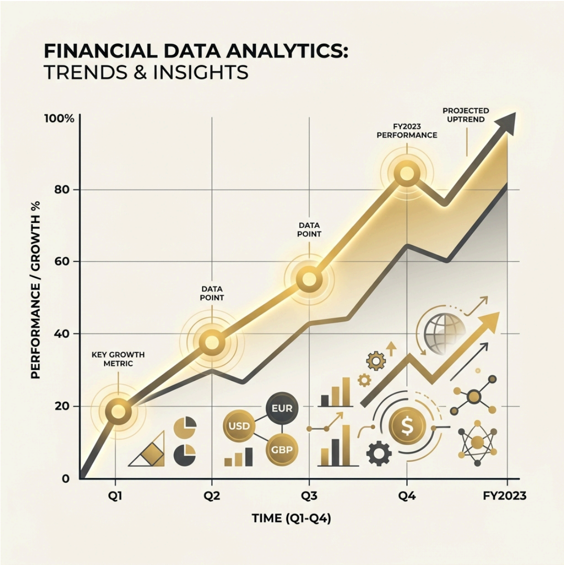
- ✓ Dense Signal - Most features carry real information and contribute to predicting target.
- ✓ Strong Multicollinearity - Input features are highly correlated with each other.
- ✓ Accuracy focus - Prioritizes keeping all parameters to squeeze out prediction error.
- ✓ Low-dimensional settings or samples outnumbering features ($N > p$).

Real Example: Housing Price Prediction

10,000 houses with 20 features (size, location, school score...). Every variable matters. Ridge shrinks all weights smoothly, outperforming OLS.

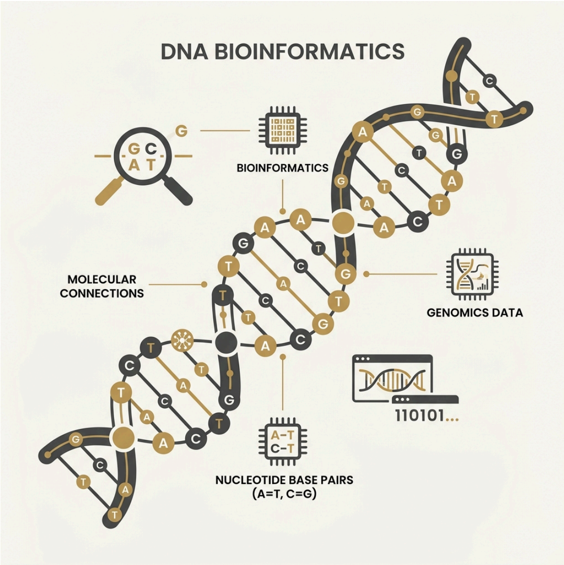
STEP 2: RESEARCH APPLICATIONS

Typical Research Fields Applying Ridge & Lasso



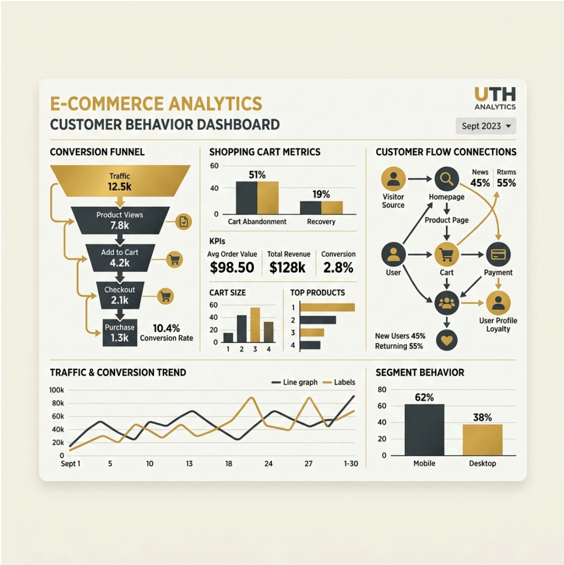
Finance & Banking

Credit scoring and stock prediction using highly collinear macro indicators. Ridge Regression stabilizes estimates without throwing away economic indicators.



Medicine & Genomics

Isolating disease-causing genetic variants where features (p ≈ 20,000) far outnumber samples (N ≈ 100). Lasso Regression prunes noise genes to exactly 0.

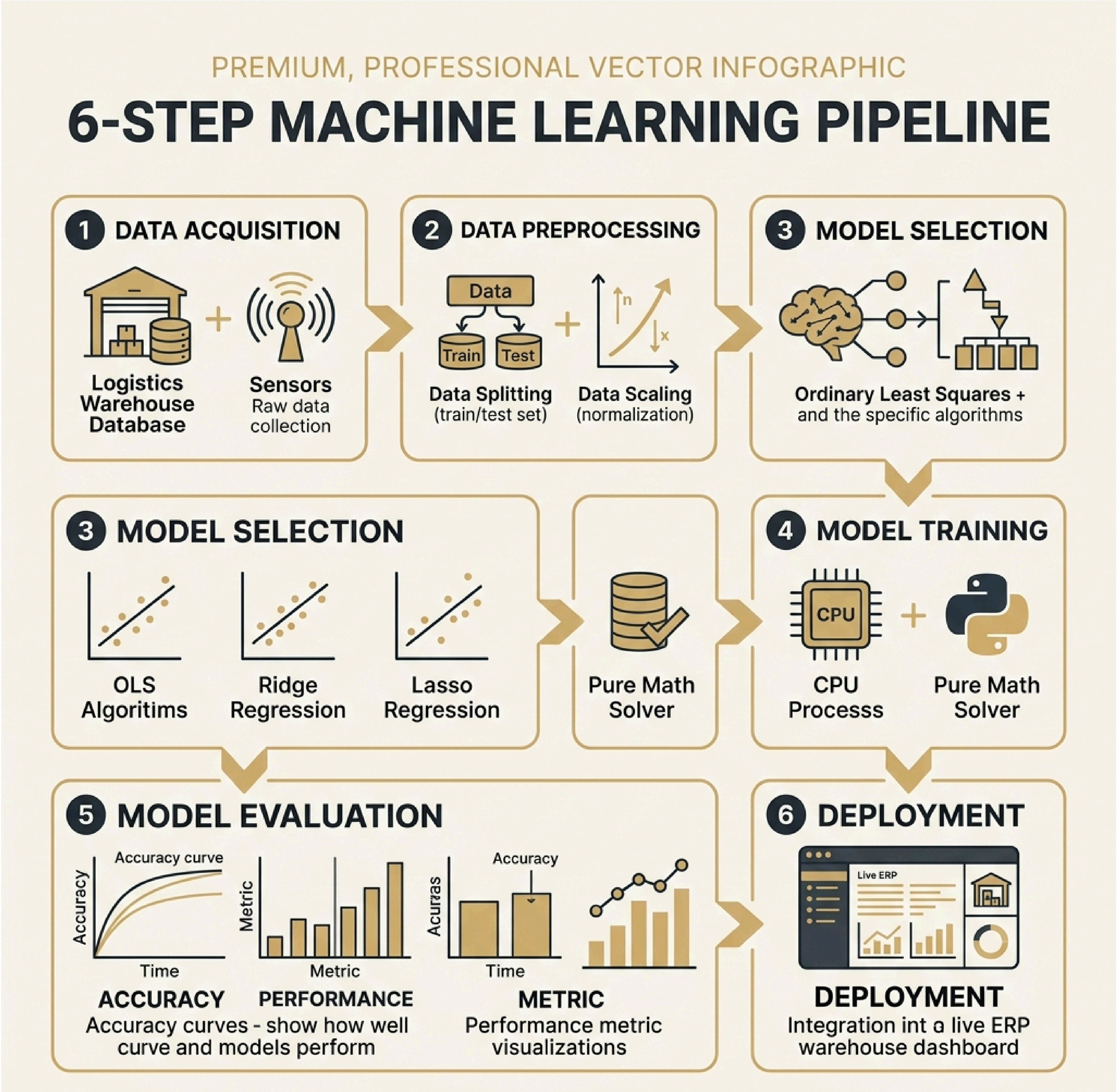


E-commerce & Marketing

Predicting Customer Lifetime Value (CLV) from noisy web interaction logs. Lasso removes irrelevant actions, creating an ultra-light real-time scoring engine.

STEP 3: CASE STUDY PIPELINE

6-Step Machine Learning Pipeline in Logistics



1. **Data Collection**

   1000 supply chain samples containing core inputs and noise temperature.

2. **Preprocessing**

   80/20 Train/Test split + Z-Score scaling + Target centering.

3. **Model Selection**

   Configuring OLS, Ridge (L2), and Lasso (L1).

4. **Model Training**

   OLS/Ridge closed-form; Coordinate Descent & Soft-Thresholding for Lasso.

5. **Evaluation**

   Validating MSE and R-squared on test samples.

6. **Deployment**

   Exporting sparse Lasso weights to ERP warehouse

STEP 4: EVALUATION & ROI

Results Analysis & Business ROI

Feature	OLS Coeff	Ridge Coeff	Lasso Coeff
Input_Qty	184.73	184.50	183.73
Output_Qty	-198.10	-197.85	-197.16
Lead_Time	21.18	21.15	20.13
Promo_Flag	9.95	9.95	8.97
Warehouse_Temp (Noise)	0.0053	-0.0070	0.0000 PRUNED

Business Value (ROI)

Lasso completely prunes temperature while keeping 99.96% accuracy. This saves hardware procurement and sensor maintenance costs.